Chapter 5 Teacher Key — Collisions Transfer Energy

> Combined answer keys: Crash Lab · Practice · Chapter 5 Quiz.

Answer Key — Crash Lab (Chapter 5)

- **Artifact:** `production/worksheets/05-ch5-crash-lab-student.docx`
- **Source alignment:** TG Lesson 11 (Investigating Collisions) · AP 11.1–11.2 · Reader pp. 25–30

Big Question

What happens when objects collide? How are collisions predictable?

Model (locked)

COLLISION → MOTION CAN CHANGE → ENERGY CAN TRANSFER → ENERGY DOES NOT DISAPPEAR

PART A — SPEED

Item	Correct/expected response	Acceptable alternatives	Expected pattern	Common misconception	What NOT to penalize
Predict	Any reasonable prediction ("the target moves farthest at 3 books").	—	stated before test	—	Not graded for correctness

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Common misconception	What NOT to penalize
Test (data table)	Target displacement recorded in cm for each ramp height (1/2/3 books), 3 trials each + average.	any plausible cm values	higher ramp → greater displacement	—	"measured speed/energy directly"
Compare	Highest ramp (3 books) moved the target farthest.	"the tallest ramp"	higher → farther	higher ramp → faster car (inferred)	"the ramp is energy"
Explain	Higher ramp → faster car → more energy of motion → greater change (target moved farther).	"faster car pushed the target farther"	speed → change	students infer speed, measure change	"force made it go"

PART B — MASS / WEIGHT

Item	Correct/ expected response	Acceptable alternatives	Expected pattern	Common misconception	What NOT to penalize
Predict	Any reasonable prediction ("3 cars moves the target farthest").	—	stated before test	—	Not graded for correctness
Test (data table)	Target displacement in cm for 1/2/3 cars, 3 trials each + average.	any plausible cm values	heavier → greater displacement	—	"measured weight of energy"
Compare	Heaviest car (3 cars) moved the target farthest.	"the heaviest"	heavier → farther	same ramp → about same speed	"heavier is always faster"
Explain	At about the same speed, a heavier car has more energy of motion → greater change.	"heavier car had more energy"	mass → change (same speed)	same speed + more mass → more energy	confound "heavier" with "faster"

PUT IT TOGETHER

Item	Correct/ expected response	Acceptable alternatives	Common misconception	What NOT to penalize
Did energy disappear?	NO — energy transferred to the target (it moved) and some became sound (+ a tiny bit of heat).	"energy moved to the cup and became sound"	"energy disappeared / was used up"	"a tiny bit of heat" optional

Notes

- Do NOT require momentum, impulse, force, contact force, or deformation terms.
- Sound = qualitative (loud/quiet) — no decibel measurement.
- Heat = not measured; "a tiny bit of heat" is optional source-aligned language.
- Reject any claim that energy "disappeared."

Variable-isolation check (the key guardrail)

- Part A: only ramp height changes (same car/target/surface). Part B: only mass changes (same ramp/target/surface).
- Any response that changes BOTH speed and mass in one comparison is a confound — redirect, do not award full credit for the "fair test" reasoning.

Answer Key — Collisions Practice (Chapter 5)

- **Artifact:** `production/worksheets/05-ch5-collisions-practice.docx`
- **Source alignment:** Reader Ch.5 pp. 25–30 · TG Lessons 10–11

Section A — WHAT CHANGED?

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
Circle changes	MOTION (speed/direction) · SHAPE · SOUND	any subset is acceptable if the example matches	collisions can change motion, shape, and produce sound	"collisions only change shape"	—
One example each	any Reader example (bowling pins = motion; clay/egg = shape; clap/drum = sound)	real-world equivalents	each change has a source example	—	informal wording

Section B — SPEED (same car)

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
B1	The fast roll moves the target farther — the same object	"faster = more energy"	same object, faster → more energy → greater	"faster car is heavier"	informal phrasing

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
	moving faster has greater energy of motion.		change		
B 2	Kept the same: the CAR, the TARGET, and the SURFACE (only speed changed).	naming all three	variable isolation	"changed everything"	listing any two of the three

Section C — MASS / WEIGHT (same speed)

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
C 1	The heavier car has more energy of motion (same speed).	"heavier = more energy"	same speed + more mass → more energy	"heavier is automatically faster"	informal phrasing
C 2	The heavier car moves the target farther.	"the heavier one"	more energy → greater change	"lighter moves farther"	—

Section D — FAIR TEST

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
D1	Change ONLY ONE thing — e.g., "change only the ramp (same car)" or "change only the car (same ramp)".	either isolated plan	one-variable-at-a-time	"change both is fine"	which single variable they choose

Section E — ENERGY DOES NOT DISAPPEAR

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
E	The car's energy of	"moved to the cup +	energy transfers,	"energy	"a tiny bit of

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	What NOT to penalize
	motion transferred to the cup (it moved) and some became sound.	became sound"	never disappears	disappeared"	heat" optional

Notes

- Reject any "energy disappeared / used up / destroyed" language.
- Do NOT require momentum/force/deformation/kinetic-potential terms.

Answer Key — Chapter 5 Quiz (Collisions Transfer Energy)

- **Artifact:** `production/quizzes/05-ch5-collisions-quiz.docx`
- **Source alignment:** Reader Ch.5 pp. 25–30 · TG Lessons 10–11 · AP 10.1/11.1/11.2 · PE 4-PS3-3

Item	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
1	b) come together with impact	—	collision definition (p.25)	"collision = disappear"	1 pt	—
2	b) it transfers	—	energy transfers	"energy	1 pt	—

#	Correct/ expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	to the still car		object→ object (p.26)	disappears"		
3	b) the target moves MORE	"farther"	same object faster → greater change (p.29)	"faster = less"	1 pt	—
4	a) the heavier car	"the heavier one"	same speed, heavier → more energy (p.30)	"lighter moves farther"	1 pt	—
5	b) sound	—	collision produces sound (p.28)	"light"	1 pt	—
6	c) it disappears	—	energy never disappears	"it transfers" (that IS what happens)	1 pt	—
7	ENERGY DOES NOT DISAPPEAR	"energy does not vanish / is not destroyed"	completes the model	leaving it blank	2 pts	wording
8	3 books (40 cm) —	"the tallest ramp"	pattern: higher ramp →	"1 book"	2 pts	informal phras

!	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	highest ramp → farthest; faster car → more energy of motion → greater change		farther			ng
9	b) change only the ramp height (same car)	—	one-variable fair test	"change everything"	1 pt	—
10	"I know the energy did not disappear because the cup moved and I heard a sound — the energy transferred to the cup and became	any change + transfer + no-disappear	change = evidence of transfer	"energy disappeared"	3 pts (claim + evidence + reasoning)	spelling; informal wording

#	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	sound."					

Out-of-scope confirmation

- No momentum · impulse · force calculations · contact force · deformation · kinetic/potential terms.
- No confounded mass+speed comparison item (items 3 and 4 each isolate one variable).
- Item 6 is the "energy disappears" distractor — the correct answer is "c) it disappears" because that is the thing that NEVER happens (the question asks what NEVER happens).

Total

- 10 items · 14 points (items 1–6, 9 = 1 pt; 7, 8 = 2 pts; 10 = 3 pts).